

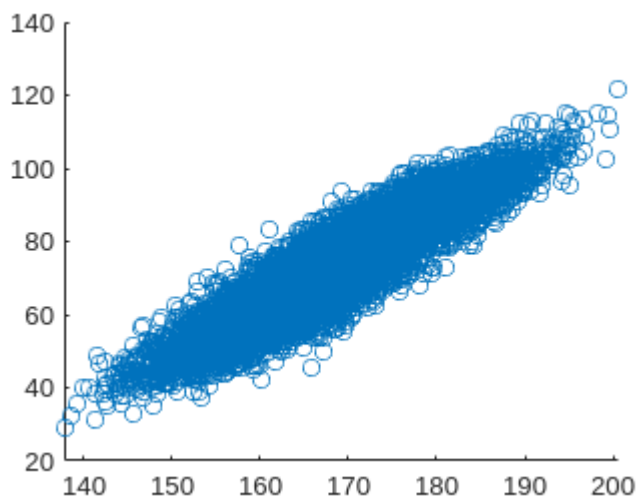
Linear Regression

Import the 'weight-height.csv' which includes 10,000 observations on 5,000 males and 5,000 females according to their **weight** (pounds) and **height** (inches). Convert the measurement into *kg* and *cm*.

```
data = readtable("weight-height.csv");  
h = data.Height*2.54;  
w = data.Weight*0.45;
```

Investigate the relationship between height and weight. Calculate the correlation.

```
scatter(h,w)
```



```
corr(h,w)
```

```
ans =  
0.9248
```

We want to estimate the parameters of the following model:

$$\text{Weight} = \beta_0 + \beta_1 \cdot \text{Height}$$

Plot the points with the regression line and 95% confidence bounds.

```
linmod = fitlm(h,w, 'linear')
```

```
linmod =  
Linear regression model:  
y ~ 1 + x1
```

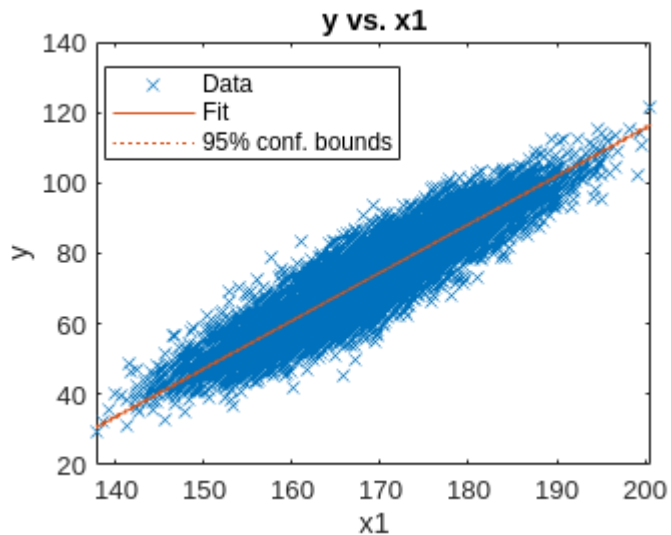
Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	-157.83	0.95017	-166.11	0
x1	1.3672	0.0056271	242.97	0

Number of observations: 10000, Error degrees of freedom: 9998

Root Mean Squared Error: 5.5
R-squared: 0.855, Adjusted R-Squared: 0.855
F-statistic vs. constant model: 5.9e+04, p-value = 0

```
plot(linmod)
```



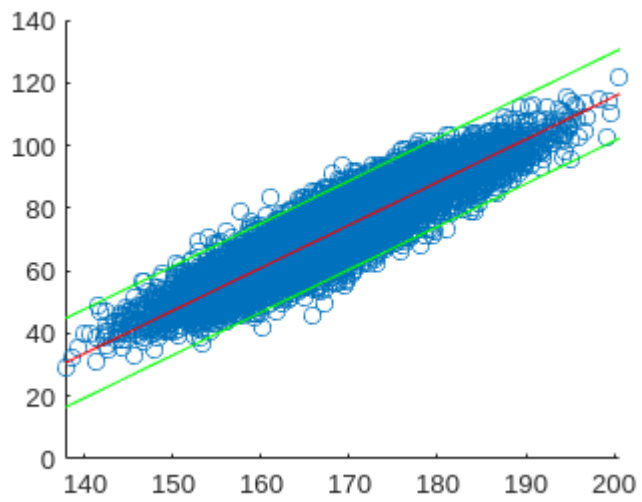
Make a 95% prediction according to your own height (cm).

```
[y ci] = predict(linmod, 198, 'Prediction', 'observation')
```

```
y =  
112.8810  
ci = 1x2  
102.0966 123.6654
```

Represent the 99% confidence bounds on the observations!

```
scatter(h,w)  
x = linspace(min(h),max(h));  
[y ci] = predict(linmod, x, 'Prediction', 'observation', ...  
    'Alpha',0.01);  
hold on  
plot(x,y, 'r',x,ci(:,1), 'g',x,ci(:,2), 'g')  
hold off
```



Exercises

1. Compare the correlations conditional to gender.
2. Use the observation related to your own gender. Compare the model parameters.